

AI Career Mentor

Innovation Sprint · Problem 1

WSU Everett · AI-Ready Workforce Certificate Program · May 15, 2026

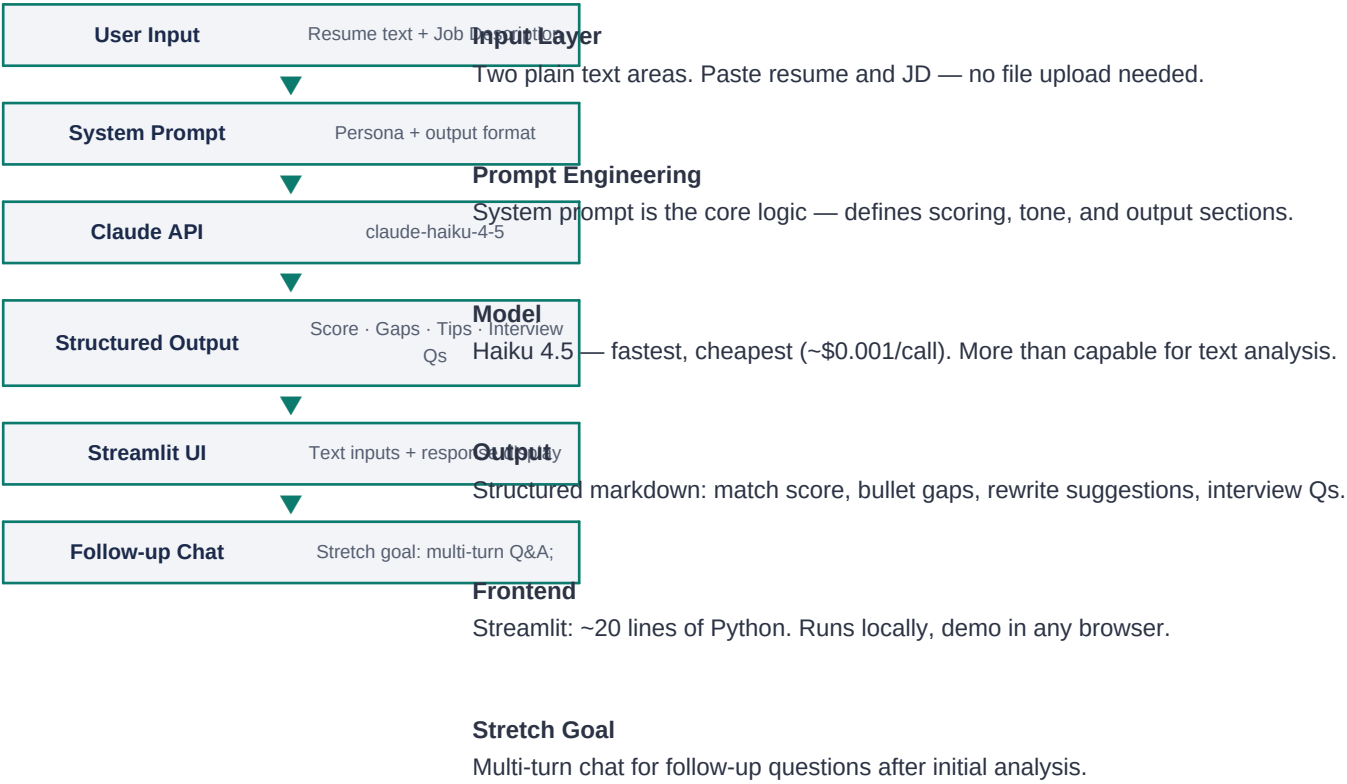
TEAM SIZE	TIME LIMIT	PROBLEM	DEMO FORMAT
5 Members	60 Minutes	Problem 1	Live Browser

01 | PROBLEM STATEMENT

Students and early-career job seekers struggle to know how well their resume matches a specific job description, what skills are missing, and how to prepare for interviews — without expensive coaching.

- Resume vs. Job Description alignment scoring
- Skill gap identification — what's missing, what's already strong
- Actionable resume improvement suggestions
- Bonus: tailored mock interview questions

02 | SYSTEM ARCHITECTURE



03 | TECH STACK

Layer	Tool	Why
AI Model	Claude Haiku 4.5 (claude-haiku-4-5-20251001)	Fast, cheap, great at structured text analysis
SDK	Anthropic Python SDK — pip install anthropic	Official library, handles auth + retries
Frontend	Streamlit — pip install streamlit	Web UI in ~20 lines of Python, no JS needed
Language	Python 3.x	Universal on the team, required by Streamlit
Secrets	ANTHROPIC_API_KEY env variable	Never hardcode the key in source files
Dataset	None required	Claude is the knowledge base — no CSV needed

04 | 60-MINUTE BUILD TIMELINE

Time	Task	Owner	Output
0–10 min	Finalize system prompt — define scoring rubric and output sections	Members 1 + 2	Locked prompt string
10–25 min	Build Streamlit UI — two text boxes, submit button, output area	Member 3	Running UI (no API yet)
10–25 min	Wire Claude API — call with resume + JD, render structured response	Member 4	Working API call
25–35 min	Integration — merge UI + API, test with a real resume and JD pair	Members 3 + 4	End-to-end prototype
35–50 min	Build 5-slide presentation deck and write talking points	Member 5	Slides ready
50–60 min	Full run-through — polish output, prep sample data, rehearse pitch	All 5	Demo-ready

05 | TEAM ROLES

Member	Role	Primary Responsibilities
Member 1	Prompt Engineer	Write + iterate system prompt, define output format, test edge cases
Member 2	AI Integration Lead	Anthropic SDK setup, API call logic, response parsing, error handling
Member 3	Frontend Dev	Streamlit layout, text inputs, submit button, output rendering
Member 4	Integration + QA	Merge UI + API, test with real resume/JD pairs, squash bugs
Member 5	Presenter + Slides	5-slide deck, talking points, leads live demo during presentation

06 | STARTER SYSTEM PROMPT

Use this as the **system** message in your API call. Prompt Engineer owns this — iterate freely.

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You are an expert career coach and resume analyst.

The user will provide:
1. Their resume (as plain text)
2. A job description (as plain text)

Return a structured response with exactly these sections:

**MATCH SCORE: X/10**
A score from 1-10 with one sentence explaining it.

**STRENGTHS**
3-4 bullets – what the resume does well for this role.

**SKILL GAPS**
3-5 bullets – skills/keywords in the JD missing from the resume.

**RESUME IMPROVEMENTS**
3-4 specific, actionable suggestions.

**MOCK INTERVIEW QUESTIONS**
2 questions the candidate should prepare for based on this JD.

Be specific, direct, and encouraging. Do not summarize the resume or JD.
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07 | PRESENTATION OUTLINE (5–7 min)

Slide	Title	Key Points
1	Team Intro	Team name, 5 members + their roles, Problem 1 chosen
2	Problem	Who suffers from this? Why resume matching matters. Cost of bad applications.
3	Solution + Architecture	Show architecture flow. Name the tools. Explain prompt-driven approach.
4	Live Demo	Paste real resume + JD live. Walk through each section of the output.
5	Future Scope	PDF upload, RAG over job boards, LinkedIn integration, interview simulator

08 | WATCH-OUTS & QUICK TIPS

- **API key first:** Add \$5 credit at `console.anthropic.com` before writing a single line of code. No credit = 401 errors on every call.
- **Environment variable:** Run `export ANTHROPIC_API_KEY='sk-ant-...'` in your terminal before launching Streamlit.
- **Model string:** Use `claude-haiku-4-5-20251001` exactly. Do not use Opus — it's 25x more expensive.
- **Demo content:** Prepare a sample resume and JD right now. Do not scramble for content during the live demo.

- **Install:** `pip install streamlit anthropic` then `streamlit run app.py`.
- **Fallback plan:** If API setup fails, build the prototype as an interactive Artifact in Claude.ai — zero install, demo directly in the browser.